

Matrix Multiplication as Composition

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Dot product ([Note 362](#)); Linear transformations and matrices ([Note 500](#)).
- **Leads to:** Jacobian and matrix gradients ([Note 602](#)).

1. Overview

Key point: Multiplying two matrices gives the matrix of “apply one transformation, then the other”; we read the product from right to left.

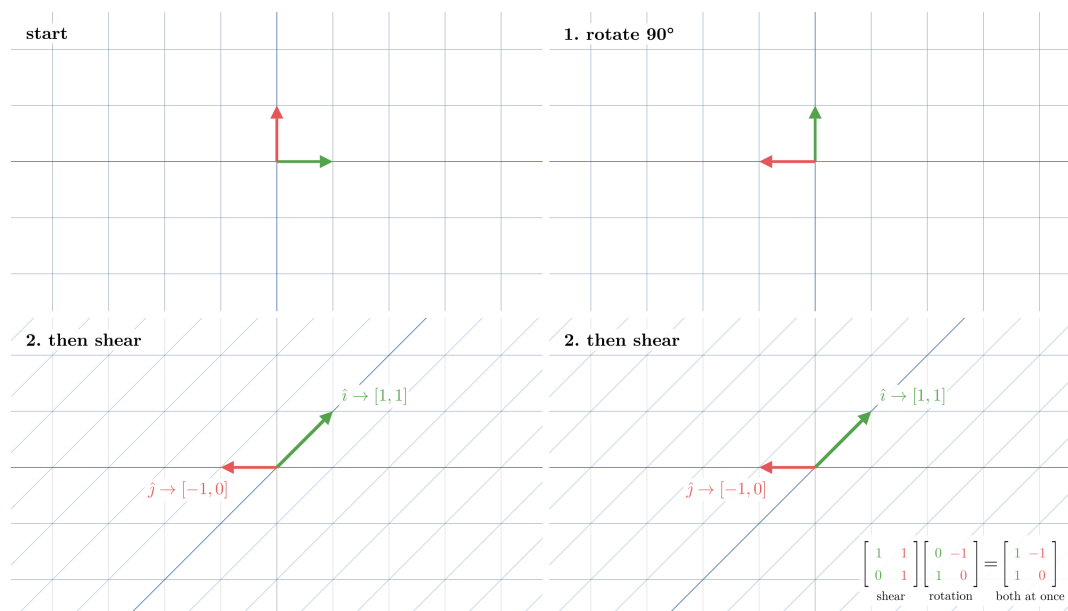


Figure 1: Rotating the plane by 90° and then shearing it is one single transformation; its matrix is the product of the two matrices

Figure 1 applies two transformations one after the other: a rotation by 90° , then a shear. From start to finish, the plane has undergone a third linear transformation, with a matrix of its own. That matrix is the product of the shear matrix and the rotation matrix.

This Note builds on the [linear transformations and matrices Note](#): the columns of a matrix are where $\hat{i}$ and $\hat{j}$ land, and $A\mathbf{x}$ is the linear combination of those columns.

2. Composition: one transformation after another

Key point: Applying one linear transformation and then another is again a linear transformation, called their composition; following $\hat{i}$ and $\hat{j}$ gives its matrix.

We often want to apply one transformation and then another. The overall effect is again a linear transformation (grid lines are still parallel and evenly spaced, and the origin has not moved). It is called the **composition** of the two transformations.

Like any linear transformation, it has a matrix, found by following $\hat{i}$ and $\hat{j}$ to their final places. In Figure 1:

- **Rotation by 90°** has the matrix R with columns $[0, 1]$ and $[-1, 0]$: it sends $\hat{i}$ to $[0, 1]$ and $\hat{j}$ to $[-1, 0]$.
- **The shear** has the matrix S with columns $[1, 0]$ and $[1, 1]$: it keeps $[1, 0]$ fixed and sends $[0, 1]$ to $[1, 1]$.
- **After both**, $\hat{i}$ has ended at $[1, 1]$ and $\hat{j}$ at $[-1, 0]$.

So the composition has the matrix with those two columns:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

This one matrix does in a single step what the rotation and the shear do in two.

3. Why the product is written right to left

Key point: In $SR\mathbf{x}$, the matrix next to the vector acts first: R is applied first, then S .

Applying the rotation and then the shear to a vector $\mathbf{x}$ the long way means two multiplications: first $R\mathbf{x}$, then S times that result, $S(R\mathbf{x})$. The composition matrix must give the same answer for every $\mathbf{x}$, since it describes the same overall movement. So it is natural to call it the **product** SR :

$$S(R\mathbf{x}) = (SR)\mathbf{x}$$

Note the order. The transformation on the right (R) happens first and the one on the left (S) second. This comes from function notation: in $f(g(x))$, g acts first. A matrix product is read from right to left.

4. Computing a product column by column

Key point: The first column of M_2M_1 is M_2 times the first column of M_1 ; the second column is M_2 times the second column of M_1 .

We can find the product without watching any animation, from the numbers alone. Take

$$M_1 = \begin{bmatrix} 1 & -2 \\ 1 & 0 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$$

and apply M_1 first, then M_2 .

1. **In words:** after M_1 , $\hat{i}$ sits at the first column of M_1 . Applying M_2 to that column gives where $\hat{i}$ finally lands: the first column of the product. Do the same with the second column for $\hat{j}$.
2. **Formula:**

$$M_2 M_1 = \left[M_2 \cdot (\text{column 1 of } M_1) \mid M_2 \cdot (\text{column 2 of } M_1) \right]$$

3. **Example:**

- $\hat{i}$ first lands on $[1, 1]$. Then $M_2[1, 1] = 1 \cdot [0, 1] + 1 \cdot [2, 0] = [2, 1]$.
- $\hat{j}$ first lands on $[-2, 0]$. Then $M_2[-2, 0] = -2 \cdot [0, 1] + 0 \cdot [2, 0] = [0, -2]$.

$$M_2 M_1 = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix}$$

The same reasoning with letters gives the general rule. Take

$$M_1 = \begin{bmatrix} e & f \\ g & h \end{bmatrix}, \quad M_2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The first column of the product is $e[a, c] + g[b, d]$ and the second is $f[a, c] + h[b, d]$:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

Extra: The same formula read entry by entry is the school rule “row times column”. The entry in row i , column j of AB is the dot product of row i of A with column j of B . In the example, row 1 of M_2 is $[0, 2]$ and column 1 of M_1 is $[1, 1]$, and $0 \times 1 + 2 \times 1 = 2$, the top-left entry. This is why the [dot product and cosine similarity Note](#) calls the dot product the building block of matrix multiplication.

Python: @ multiplies two matrices.

```
import numpy as np

M1 = np.array([[1, -2], [1, 0]])
M2 = np.array([[0, 2], [1, 0]])
M2 @ M1           # [[2, 0], [1, -2]]
x = np.array([1, 1])
M2 @ (M1 @ x)     # [2, -1]: two steps
(M2 @ M1) @ x     # [2, -1]: one step
```

5. Order matters

Key point: In general $AB \neq BA$: shearing then rotating is a different transformation from rotating then shearing.

Does the order of the two matrices matter? Thinking in transformations, we can answer by picturing them (Figure 2).

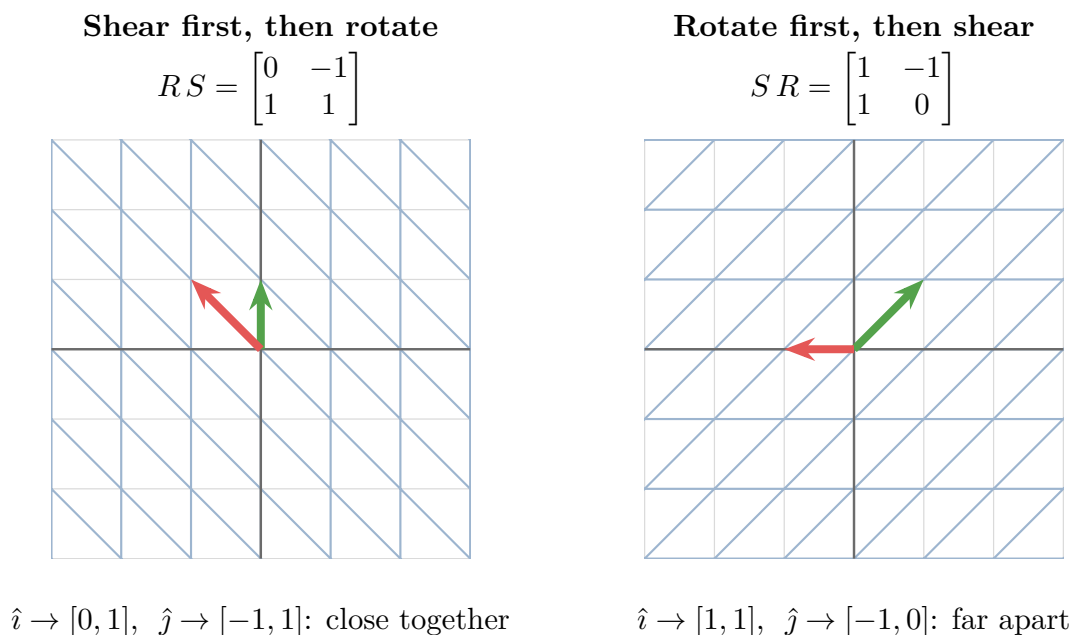


Figure 2: The same two transformations in the two possible orders

- **Shear first, then rotate** (RS): $\hat{i}$ ends at $[0, 1]$ and $\hat{j}$ at $[-1, 1]$. The two vectors point close together.
- **Rotate first, then shear** (SR): $\hat{i}$ ends at $[1, 1]$ and $\hat{j}$ at $[-1, 0]$. They point far apart.

The overall effects differ, so $RS \neq SR$:

$$RS = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} = SR$$

Matrix multiplication is **not commutative**, unlike the multiplication of numbers. In code this means $A @ B$ and $B @ A$ are, in general, different matrices.

6. Grouping does not matter

Key point: $(AB)C = A(BC)$, because both mean “apply C , then B , then A ”.

For three matrices A , B and C , we can first compute AB and multiply the result by C , or first compute BC and put A in front. Both give the same matrix:

$$(AB)C = A(BC)$$

This property is called **associativity**. Proving it by expanding all the entries is long and tells us nothing. Seen as transformations it is immediate: both sides mean “apply C , then B , then A ”, the same three steps in the same order. Only the bookkeeping differs.

Python: A quick check with the three matrices of this Note.

```
R = np.array([[0, -1], [1, 0]])
S = np.array([[1, 1], [0, 1]])
np.array_equal((S @ R) @ M1, S @ (R @ M1))    # True
np.array_equal(S @ R, R @ S)                  # False
```

7. Where ML multiplies matrices

Key point: A product needs matching inner sizes; stacked linear layers collapse into one matrix, which is why neural networks need activation functions between them.

7.1 The shape rule

Key point: An $m \times n$ matrix times an $n \times p$ matrix gives an $m \times p$ matrix; the inner sizes must match.

So far all matrices were 2×2 . In ML the matrices are rectangular, and the column-by-column view explains the shape rule. In AB , each column of B is a vector that A transforms, so each column of B must have as many entries as A has columns.

1. **In words:** the number of columns of the left matrix must equal the number of rows of the right matrix; the result takes the rows of the left and the columns of the right.
2. **Formula:**

$$(m \times n)(n \times p) = (m \times p)$$

3. **Example:** the data matrix X with 3 points and 2 features times A^T in the [linear transformations and matrices Note](#) (section 7.1) is $(3 \times 2)(2 \times 2) = 3 \times 2$: three transformed points. PCA’s XW^T with 40 points, 3 features and 2 components is $(40 \times 3)(3 \times 2) = 40 \times 2$ (see the [PCA step by step Note](#), section 5.1).

The dot product $a^T b$, $(1 \times n)(n \times 1) = 1 \times 1$, is the smallest case of this rule.

7.2 Linear layers collapse into one

Key point: Two layers without an activation, $W_2(W_1 \mathbf{x})$, are the single layer $(W_2 W_1) \mathbf{x}$: stacking them adds no power.

A neural network applies layer after layer. Suppose two layers had no activation function between them, only the matrices W_1 and then W_2 . By Section 3, the two steps are one transformation:

$$W_2(W_1\mathbf{x}) = (W_2W_1)\mathbf{x}$$

With the numbers of Section 4, $W_1 = M_1$ and $W_2 = M_2$: the input $[1, 1]$ goes to $M_1[1, 1] = [-1, 1]$ and then to $M_2[-1, 1] = [2, -1]$. The single matrix M_2M_1 sends $[1, 1]$ straight to $[2, -1]$. However many linear layers we stack, the result is still one matrix, one linear transformation.

This is why every layer of a network ends with a non-linear activation such as the [sigmoid](#). The activation bends the space between layers, so the composition can no longer be squeezed into a single matrix, and depth adds real power.

Extra: The same collapse shows in linear regression with engineered features. If every new feature is a linear combination of the old ones, the model can learn nothing new: the combined effect is still a linear combination of the original columns. Polynomial features (see the [polynomial regression Note](#)) help precisely because squaring is not linear.

8. Summary

Idea	What it says	Example
Composition	one transformation after another	rotate, then shear
Product BA	matrix of “apply A , then B ”	$SR = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$
Column by column	column j of BA is B times column j of A	$M_2[1, 1] = [2, 1]$
Not commutative	$AB \neq BA$ in general	$RS \neq SR$
Associative	$(AB)C = A(BC)$	same three steps, same order
Shape rule	$(m \times n)(n \times p) = m \times p$	$(40 \times 3)(3 \times 2) = 40 \times 2$

- A matrix product is a composition of transformations, read from right to left.
- Follow $\hat{i}$ and $\hat{j}$ through both steps to get the columns of the product.
- Order matters; grouping does not.
- Stacked linear layers collapse into one matrix; activation functions prevent that.

9. Key terms

Term	Meaning
Composition	Applying one transformation and then another, seen as one overall transformation
Matrix product	The matrix BA of the composition “apply A , then B ”
Not commutative	The order of the factors matters: $AB \neq BA$ in general
Associativity	$(AB)C = A(BC)$: the grouping of a product does not matter
Shape rule	$(m \times n)(n \times p) = m \times p$; the inner sizes must match